Smooth Manifolds

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Abstract

Contents

1 Basic Properties of Smooth Manifolds

1.1 A Sufficient Criterion for Smooth Submanifolds

Proposition 1.1.1
Let M be a smooth manifold of dimension n. Let $f = (f_1, \dots, f_m) : M \to \mathbb{R}^m$ be a smooth map. Let $p \in f^{-1}(0)$. Let (U, ϕ) be a chart at p. If the matrix

$$\begin{pmatrix} \frac{\partial(f_1\circ\phi^{-1})}{\partial x_1}|_{\phi(p)} & \cdots & \frac{\partial(f_1\circ\phi^{-1})}{\partial x_n}|_{\phi(p)} \\ \vdots & \ddots & \vdots \\ \frac{\partial(f_m\circ\phi^{-1})}{\partial x_1}|_{\phi(p)} & \cdots & \frac{\partial(f_m\circ\phi^{-1})}{\partial x_n}|_{\phi(p)} \end{pmatrix}$$

has rank m, then $f^{-1}(0)$ is a smooth submanifold of dimension $n - m$.

Proof
Notation change: Write $\frac{\partial f_i}{\partial x^j}$ in place of $\frac{\partial(f_i\circ\phi^{-1})}{\partial x_j}$.

Since the given matrix M has rank n, we can relabel the f_i and the x_j so that the first $n \times n$ block matrix $\left(\frac{\partial f_i}{\partial x^j}\right)_{1\leq i,j\leq n}$ has non-zero determinant. In other words, our $m \times n$ matrix is now of the form

$$\begin{pmatrix} \left(\frac{\partial f_i}{\partial x^j}\right)_{1\leq i,j\leq n} & * \end{pmatrix} \in \text{Mat}_{m\times n}$$

Now notice that $Y = f^{-1}(0)$ is our set in question. Let $p \in Y$. Let $(U, \phi = (x^1, \dots, x^m))$ be an arbitrary choice of chart at p on M. Then notice that $(U, (f_1, \dots, f_n, x^{n+1}, x^m))$ is also a chart on M. Indeed $f_1, \dots, f_n$ are assumed to be smooth so that it is compatible with the atlas of M. Now the Jacobian of ϕ^{-1} composed of this chart, which is an $m \times m$ matrix, is given by the formula

$$\begin{pmatrix} \left(\frac{\partial f_i}{\partial x^j}\right)_{1\leq i,j\leq n} & * \\ 0 & I \end{pmatrix} \in \text{Mat}_{m\times m}$$

Its determinant is precisely

$$\left|\left(\frac{\partial f_i}{\partial x^j}\right)_{1\leq i,j\leq n}\right|$$

which is non-zero by assumption. Then the inverse function theorem applies.

In the chart $(U, (f_1, \dots, f_n, x^{n+1}, \dots, x^m))$, the set $Y = f^{-1}(0)$ is obtained by setting the the vanishing of the first m coordinate functions $f_1, \dots, f_n$. Thus $\phi(U \cap Y)$ is now equal to $\phi(U) \cap \mathbb{R}^{m-n}$ so that Y is an embedded submanifold. ■

Example 1.1.2
The set $N = \{(x, y) \in \mathbb{R}^2 \mid y^2 - x^2 + x = 0\}$ is a smooth submanifold of $\mathbb{R}^2$.

Proof
We compute the matrix of the function $f(x, y) = y^2 - x^2 + x$ to get $(-2x + 1, 2y)$. This does not have rank 1 if and only if $x = \frac{1}{2}$ and $y = 0$. But this point lies outside of N. Hence by the above prp, N is a smooth submanifold of $\mathbb{R}^2$. ■

Note that this is a slightly weaker statement of the preimage theorem 4.3.3. In that theorem, instead of using the C^∞ map $F = (f_1, \dots, f_n)$ that maps to $\mathbb{R}^n$, it considers in what situation will a smooth map of manifolds $f : M \to N$ give a submanifolds on a level set $f^{-1}(c)$ for $c \in N$.

1.2 Partitions of Unity

Smooth partitions of unity is unique to manifold that are locally $\mathbb{R}^n$. Complex manifolds would have a more rigid structure in the sense that they do not have a partition of unity. In our case of smooth manifolds, we can construct a smooth partition of unity. This allows us to piece together functions defined on charts on a manifold.

Definition 1.2.1 (Support of a Function) Let $f : M \to \mathbb{R}$ be a function on a manifold M. Define the support of f to be

$$\text{supp}(f) = \overline{\{p \in M | f(p) \neq 0\}}$$

Definition 1.2.2 (Compact Exhaustion) Let X be a topological space. A compact exhaustion of X is a sequence of open subsets $\{K_i | i \in \mathbb{N}\}$ such that $K_0 \subseteq K_1 \subseteq K_2 \subseteq \cdots$ such that each $\overline{K_i}$ is compact, $\overline{K_i} \subset K_{i+1}$ and $X = \bigcup_{i=1}^\infty K_i$.

Proposition 1.2.3 Every manifold M admits a compact exhaustion.

Proof Firstly, since M has a countable basis from second countability, and M is locally Euclidean, we can find a countable basis for the topology of M, namely $\{V_j | j \in \mathbb{N}\}$ such that each V_j has compact closure. Indeed, we can just throw out any V_j from the original list that does not have compact closure. Set $G_1 = V_1$. Suppose inductively that $G_k = \bigcup_{i=1}^{j_k} V_i$. Let j_{k+1} be the smallest positive integer greater than j_k such that $\overline{G_k} \subset \bigcup_{i=1}^{j_{k+1}} V_i$ and put $G_{k+1} = \bigcup_{i=1}^{j_{k+1}} V_i$. This gives a compact exhaustion as required. ■

Definition 1.2.4 (Partitions of Unity) A partition of unity on a manifold M is a collection of smooth non-negative functions $\{\theta_i : M \to \mathbb{R} | i \in I\}$ such that the following are true.

- For all $p \in M$ and $i \in I$, $0 \leq \theta_i(p) \leq 1$
- For each $p \in M$, there is a neighbourhood U of p such that there is only a finite number of θ_i with non-empty support.
- $\sum_{i \in I} \theta_i = 1$

Definition 1.2.5 (Subordinate Partitions) We say that a partition of unity $\{\theta_i : M \to \mathbb{R} | i \in I\}$ for a manifold M is subordinate to an open cover $\{U_\alpha | \alpha \in A\}$ of M if $\text{supp}(\theta_i) \subset U_\alpha(i)$ for every $i \in I$.

Theorem 1.2.6 (Existence of a Partition of Unity) Let M be a manifold and $\{U_\alpha | \alpha \in A\}$ an open cover of M. Then there exists a partition of unity subordinate to the open cover $\{U_\alpha | \alpha \in A\}$.

Proof We first prove a result in $\mathbb{R}^n$. Namely: Let $C(r) = \{(x_1, \dots, x_n) \in \mathbb{R}^n || x_i| < r\}$ be the open cube of side length $2r$ centered at the origin. Then there exists a non-negative smooth function φ on $\mathbb{R}^n$ such that $\varphi(x) = 1$ for $x \in \overline{C(1)}$ and $\varphi(x) = 0$ for $x \in \mathbb{R}^n \setminus C(2)$. We just have to do it in dimension $n = 1$ since we can just take product of such functions. One possible way to write down a formula is: Start with

$$f(t) = \begin{cases} e^{-1/t} & \text{if } t > 0 \\ 0 & \text{if } t \leq 0 \end{cases}$$

Put $g(t) = \frac{f(t)}{f(1)+f(1-t)}$ which is non-negative, smooth and equal to 1 for $t \geq 1$ and 0 for $t \leq 0$. Then setting $\varphi(t) = g(t+2)g(2-t)$ gives the function as required.
Now for the main proof. Construct a compact exhaustion for M and set $G_0 = \emptyset$. For a point $p \in M$, let i_p be the largest integer such that $p \in M \setminus \overline{G_{i_p}}$. We also pick an index $\alpha_p \in A$ such that $p \in U_{\alpha_p}$. Now let (V, τ) be a chart around p with $\tau(p) = 0$ and such that

$$V \subset U_{\alpha_p} \cap (G_{i_p+2} \setminus \overline{G_{i_p}})$$

and such that $\tau(V)$ contains the closed cube $\overline{C(2)}$. Now define

$$\psi_p(q) = \begin{cases} \varphi \circ \tau(q) & \text{if } q \in V \\ 0 & \text{otherwise} \end{cases}$$

where φ is the smooth function constructed above. Clearly ψ_p is a smooth function on M taking the value 1 on some open neighbourhood W_p of p with compact support contained in $V \subset U_{\alpha_p} \cap (G_{i_p+2} \setminus \overline{G_{i_p}})$.
For each $i \geq 1$, choose a finite set of points $p \in M$ whose neighbourhoods W_p as above cover $\overline{G}_i \setminus G_{i-1}$. This is possible since the $\overline{G}_i \setminus G_{i-1}$ is compact. Order all functions ψ_p corresponding to such points p in a sequence

$$\psi_1, \psi_2, \psi_3, \dots$$

Clearly, for every point in M, only finitely many of the ψ_j are nonzero at that point since the set of ψ with support contained in each G_i is finite by construction. This means that

$$\psi = \sum_{j=1}^{\infty} \psi_j$$

is a well defined smooth function on M with $\psi(p) > 0$ everywhere. Then define

$$\theta_i = \frac{\psi_i}{\psi}$$

Then these θ_i have all the required properties. ∎

This tool will be useful for us to patch up local expression and normalize it so that the output is a globally defined object on the manifold, provided that they agree on intersections.

1.3 The Whitney Embedding Theorem

Theorem 1.3.1 (The (Strong) Whitney Embedding Theorem) Any smooth manifold of dimension m can be smoothly embedded into $\mathbb{R}^{2m}$.

2 The Tangent Space

2.1 Tangent Space

Conventionally there are two ways of defining a tangent vector on a smooth manifold. We will take the more geometric approach by considering curves on the manifold. To begin, we need some algebra.

Definition 2.1.1 ($\mathbb{R}$-Algebra of Germs of $\mathcal{C}^\infty$ Functions) Let M be a manifold. Let $p \in M$. Define the $\mathbb{R}$-Algebra of Germs of Functions

$$\mathcal{C}^\infty_{M,p} = \{(f : U \to \mathbb{R}, U) \mid p \in U, f \in \mathcal{C}^\infty\}/\sim$$

where we say that $(f, U) \sim (g, V)$ if there exists some open neighbourhood W of p so that $W \subset U \cap V$ and $f|_W = g|_W$.

Since we are only concerned with tangents of the curves in order to form the tangent space, whenever two functions coincide on some open subset, they will have the same tangents there. So indeed the defintion makes sense. However, two different germs can still have the same tangent vector. This appear say for example when comparing the tangent vector at 0 of the function $f(x) = 0$ and $g(x) = x^2$.

Readers with algebraic geometry background will notice that this is very similar to the definition of sheaves. Indeed it is a sheaf of $\mathbb{R}$-algebra over M. Moreover, this germs are elements of the stalk of a sheaf.

Definition 2.1.2 (Equivalence Class of Curves) Let M be a smooth manifold of dimension n. Let $p \in M$ and (U, ϕ) a coordinate chart on p. We say that two curves $\gamma_1, \gamma_2 : (-1, 1) \to \mathbb{R}^n$ are equivalent if

$$\left.\frac{d(\phi \circ \gamma_1)(t)}{dt}\right|_0 = \left.\frac{d(\phi \circ \gamma_2)(t)}{dt}\right|_0$$

they have the same derivative at 0 under the coordinate chart.

It is easy to check that this equivalence class is not affected by the choice of the coordinate chart.

Definition 2.1.3 (Tangent Vectors) Let M be a smooth manifold. Let $p \in M$ and $[\gamma]$ an equivalence class of curves on M such that $\gamma(0) = p$. Define the tangent vector to the curve γ at p to be the map $X_{\gamma,p} : \mathcal{C}^\infty_{M,p} \to \mathbb{R}$ defined by

$$X_{\gamma,p}(f) = \left.\frac{d(f \circ \gamma)(t)}{dt}\right|_{t=0}$$

Notice that this construction shows that tangent vectors are elements of the dual space of $\mathcal{C}^\infty_{M,p}$. This will be called the cotangent space in later sections, and it appears more naturally then tangent space. Moreover, the definition varies with γ instead of anything else. By allowing γ to be different curves, we obtain different tangent vectors. In particular, changing the germ f does not lead to new tangents: it only modifies the height of the output as well as the domain.

It is easy to see that two different representatives of the same equivalence class of curves give the same tangent vector.

What follows is that we will show that these are essentially the same construction, where we have the tangent vectors being an explicit construction.

Definition 2.1.4 (Tangent Space) Let M be a manifold. Define the tangent space to M at p to be the set of all tangent vectors of M at p, denoted $T_p(M)$.

Definition 2.1.5 (Standard Basis Vectors of the Tangent Space With Respect to a Chart)
Let M be a smooth manifold. Let $p \in M$. Let $(U, \phi = (x^1, \dots, x^n))$ be a chart containing p. Define the tangent vector $\frac{\partial}{\partial x^k}\Big|_p : \mathcal{C}^\infty_{M,p} \to \mathbb{R}$ by

$$\frac{\partial}{\partial x^k}\Big|_p (f) = \frac{\partial f \circ \phi^{-1}}{\partial x_k}\Big|_{\phi(p)}$$

Proposition 2.1.6 (Standard Basis)
Let M be a smooth manifold. Let $p \in M$. Let $(U, \phi = (x^1, \dots, x^n))$ be a chart containing p. Then an $\mathbb{R}$-basis for $T_p(M)$ is given by

$$\left\{ \frac{\partial}{\partial x^1}\Big|_p, \dots, \frac{\partial}{\partial x^n}\Big|_p \right\}$$

Proof By chain rule applied to the map $(f \circ \phi^{-1}) \circ (\phi \circ \gamma) : \mathbb{R} \to \mathbb{R}^n$, for any $\gamma : (-1, 1) \to M$ and $f \in \mathcal{C}^\infty_{M,p}$, we have

$$X_{\gamma,p}(f) = \sum_{k=1}^n \frac{\partial (f \circ \phi^{-1})}{\partial x_k}\Big|_{\phi(p)} \cdot \frac{d\gamma_k(t)}{dt}\Big|_{t=0}$$

where $\gamma_k(t) = x^k$. Rewriting $a_k(p) = \frac{d\gamma_k(t)}{dt}\Big|_{t=0}$, we have that

$$X_{\gamma,p}(f) = \sum_{k=1}^n a_k \frac{\partial (f \circ \phi^{-1})}{\partial x_k}\Big|_{\phi(p)}$$

By allowing f to vary, we obtain $X_{\gamma,p} = \sum_{k=1}^n a_k \frac{\partial}{\partial x^k}\Big|_p$. It remains to show that they span the vector space and that they are linearly independent.

On one hand, given a linear combination $\sum_{k=1}^n a_k \frac{\partial}{\partial x^k}\Big|_p$, let $(U, \phi = (x^1, \dots, x^n))$ be a chart at p and consider the curve $\gamma : (-1, 1) \to M$ given on local charts by

$$(\phi \circ \gamma)(t) = \begin{pmatrix} x^1(p) + a_1(p)t \\ \vdots \\ x^n(p) + a_n(p)t \end{pmatrix}$$

Then the tangent vector to the curve γ at p is given by

$$\begin{aligned} X_{\gamma,p}(f) &= \frac{d(f \circ \gamma)(t)}{dt}\Big|_{t=0} \\ &= \left(\frac{\partial f \circ \phi^{-1}}{\partial x_1}\Big|_{\phi(p)} \quad \cdots \quad \frac{\partial f \circ \phi^{-1}}{\partial x_n}\Big|_{\phi(p)} \right) \cdot \frac{d(\phi \circ \gamma)(t)}{dt}\Big|_{t=0} \\ &= \sum_{k=1}^n a_k(p) \frac{\partial f \circ \phi^{-1}}{\partial x_k}\Big|_{\phi(p)} \end{aligned}$$

for any choice of $f \in C^\infty_{M,p}$. On the other hand, each of $\frac{\partial}{\partial x^k}\Big|_p$ are linearly independent because if

$$\sum_{k=1}^n a_k(p) \frac{\partial}{\partial x^k}\Big|_p = 0$$

then we have that

$$0 = \sum_{k=1}^n a_k(p) \frac{\partial x^i}{\partial x^k}\Big|_p = a_k(p)$$

so that $\left\{ \frac{\partial}{\partial x^1}\bigg|_p, \dots, \frac{\partial}{\partial x^n}\bigg|_p \right\}$ is a linearly independent set. ■

This means that a tangent vector of M has the form

$$X_{\gamma,p} = \sum_{k=1}^n a_k(p) \frac{\partial}{\partial x^k}\bigg|_p \quad \text{and} \quad X_{\gamma,p}(f) = \sum_{k=1}^n a_k(p) \frac{\partial(f \circ \phi^{-1})}{\partial x_k}\bigg|_{\phi(p)}$$

where $a_k(p) \in \mathbb{R}$ depends on $[\gamma]$ for each k.

This proposition also shows that the choice of γ becomes encoded in the coefficients of the basis elements $a_1(p), \dots, a_n(p) \in \mathbb{R}$. Thus from now on we omit γ in $X_{\gamma,p}$ and simply write X_p. Later we will see that as we smoothly vary the point p, if each $a_1, \dots, a_n$ varies smoothly with points on M so that $a_1, \dots, a_n \in C^\infty_{M,p}$, then X becomes a vector field.

Corollary 2.1.7
Let M be a smooth manifold. Then we have $\dim_{\mathbb{R}}(T_p(M)) = \dim(M)$.

Given two charts overlapping on a point $p \in M$, how does the derivative of the functions change with respect to changing their coordinate charts? The following proposition shows that once again, the question can be reduced to a problem on multivariable calculus.

Proposition 2.1.8
Let M be a smooth manifold. Let $p \in M$. Let $(U, \phi = (x^1, \dots, x^n))$ and $(V, y^1, \dots, y^n)$ be two chart containing p. Then

$$\frac{\partial}{\partial x^k}\bigg|_p = \sum_{i=1}^n \frac{\partial(y^i \circ \phi^{-1})}{\partial x_k}\bigg|_p \frac{\partial}{\partial y^k}\bigg|_p$$

on $U \cap V$.

2.2 Equivalent Characterizations of the Tangent Space

Lemma 2.2.1
Let M be a smooth manifold. Then

$$m_p = \{f \in \mathcal{C}^\infty_{M,p} \mid f(p) = 0\}$$

is a maximal ideal of $\mathcal{C}^\infty_{M,p}$.

Proposition 2.2.2
Let M be a smooth manifold. Then the following are true.

- There is a natural $\mathbb{R}$-linear isomorphism

$$T_p(M) \cong \text{Der}_{\mathbb{R}}(\mathcal{C}^\infty_{M,p}, \mathbb{R})$$

  given by $[\gamma] \mapsto X_{\gamma,p}$ where $\mathbb{R}$ is a $\mathcal{C}^\infty_{M,p}$-module via the isomorphism $\mathbb{R} \cong \frac{\mathcal{C}^\infty_{M,p}}{m_p}$.
- There is a natural $\mathbb{R}$-linear isomorphism

$$\text{Der}_{\mathbb{R}}(\mathcal{C}^\infty_{M,p}, \mathbb{R}) \cong \left(\frac{m_p}{m_p^2}\right)^*$$

  given by restriction of a derivation d to the ideal m_p and then taking quotient.

2.3 Differential of Smooth Maps

Definition 2.3.1 (Differential of a Map)
Let M, N be smooth manifolds. Let $f : M \to N$ be a smooth map. Let $p \in M$. Define the differential of f at p to be the map $(f_*)_p : T_p(M) \to T_p(N)$ given by

$$(f_*)_p(X_p) = X_p(- \circ f)$$

Theorem 2.3.2
Let M, N be smooth manifolds. Let $f : M \to N$ be a smooth map. Let $p \in M$. Let $(U, x^1, \dots, x^n)$ be a chart of M containing p. Let $(V, y^1, \dots, y^m)$ be a chart of N containing $f(p)$. Then $(f_*)_p$ is given by the matrix that is the Jacobian:

$$(f_*)_p = \begin{pmatrix} \frac{\partial}{\partial x^1}\Big|_p (y^1 \circ f) & \cdots & \frac{\partial}{\partial x^n}\Big|_p (y^1 \circ f) \\ \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x^1}\Big|_p (y^m \circ f) & \cdots & \frac{\partial}{\partial x^n}\Big|_p (y^m \circ f) \end{pmatrix}$$

To compute the differential of a map applied to a tangent vector, write $X_{\gamma,p}$ in terms of the basis at (U, ϕ). For example, $X_{\gamma,p}$ can be given by $\begin{pmatrix} a_1(p) & \cdots & a_n(p) \end{pmatrix}^T$. Then apply the linear transformation $(f_*)_p$ to receive an $m \times 1$ vector:

$$(f_*)_p(X_{\gamma,p}) = \begin{pmatrix} \frac{\partial(y_1 \circ f \circ \phi^{-1})}{\partial x_1}|_{\phi(p)} & \cdots & \frac{\partial(y_1 \circ f \circ \phi^{-1})}{\partial x_n}|_{\phi(p)} \\ \vdots & \ddots & \vdots \\ \frac{\partial(y_m \circ f \circ \phi^{-1})}{\partial x_1}|_{\phi(p)} & \cdots & \frac{\partial(y_m \circ f \circ \phi^{-1})}{\partial x_n}|_{\phi(p)} \end{pmatrix} \begin{pmatrix} a_1(p) \\ \vdots \\ a_n(p) \end{pmatrix}$$

The resulting vector is then the tangent vector of $T_{f(p)}(N)$ in terms of the basis elements around the chart of N.

Proposition 2.3.3 (Chain Rule) Let $f : M \to N$ and $g : N \to P$ be a smooth map on manifolds M, N, P. Let $p \in M$. Then

$$((g \circ f)_*)_p = (g_*)_{f(p)} \circ (f_*)_p$$

where the differential is taken as the tangent space over p, $f(p)$ and $g(f(p))$.

The following corollary will use the fact that the differential of the identity map is again the identity. This can be seen using the Jacobian description of the differential.

Corollary 2.3.4 Let $f : M \to N$ be a diffeomorphism of smooth manifolds M and N. Then for any $p \in M$, $(f_*)_p : T_p(M) \to T_{f(p)}(N)$ is an isomorphism of vector spaces.

Proof If $f : M \to N$ is a diffeomorphism, then f has a smooth inverse f^{-1}. Then

$$\mathrm{id}_{T_p(M)} = (\mathrm{id}_*)_p = ((f^{-1} \circ f)_*)_p = ((f^{-1})_*)_{f(p)} \circ (f_*)_p$$

and similarly for $\mathrm{id}_{T_q(N)} = (f_*)_{f^{-1}(q)} \circ ((f^{-1})_*)_q$ show that $(f_*)_p$ is an isomorphism. ∎

The chain rule together with the fact that $(\mathrm{id}_*)_p = \mathrm{id}_{T_p(M)}$ show that the map $f \mapsto (f_*)_p$ is functorial for any choice of $p \in M$.

Using the differential, we can give an alternate proof for the basis of the vector space $T_p(M)$.

For each $\phi : U \to \mathbb{R}^n$ a chart on M, ϕ is a diffeomorphism onto its image so that the differential ϕ_* :

$T_p(M) \to T_{\phi(p)}(\mathbb{R}^n)$ is an isomorphism. In particular, the tangent vector $\frac{\partial}{\partial x^k}\Big|_p$ is sent to

$$\phi_*\left(\frac{\partial}{\partial x^k}\Big|_p\right)(g) = \frac{\partial(g \circ \mathrm{id}_{\phi(U)})}{\partial x_k}\Big|_{\phi(p)} = \frac{\partial g}{\partial x_k}\Big|_{\phi(p)}$$

for any choice $g \in C^\infty_{\mathbb{R}^n,\phi(p)}$. Notice that the differential changed operator because x^k refers to coordinates on the manifold while x_k refers to coordinates on $\mathbb{R}^n$. Now the isomorphism ϕ_* carries basis to basis and in multivariable calculus we have already seen that elements of the latter form a basis for $T_p(\mathbb{R}^n)$.

2.4 The Preimage Theorem

Definition 2.4.1 (Regular Values) Let $f : M \to N$ be a smooth map of smooth manifolds. We say that $q \in N$ is a regular value if for $p \in f^{-1}(q)$, $(f_*)_p$ is surjective. Otherwise it is said to be a critical value.

Definition 2.4.2 (Regular Level Set) Let $f : M \to N$ be a smooth map of smooth manifolds. We say that a level set $f^{-1}(q)$ for $q \in N$ is regular if for all $p \in f^{-1}(q)$, p is a regular value.

The preimage theorem is a variation of the inverse function theorem, except that it is applied to a manifold instead of $\mathbb{R}^n$.

Theorem 2.4.3 (The Preimage Theorem) Let $f : M \to N$ be a smooth map of smooth manifolds where $\dim(M) = m$ and $\dim(N) = n$. Let $q \in N$. If $f^{-1}(q)$ is a non empty regular level set, then $f^{-1}(q)$ is an embedded submanifold of N of dimension $m - n$.

Note the similarities of the preimage theorem and proposition 2.2.3. Proposition 2.2.3 is actually a special case of the preimage theorem. By choosing $N = \mathbb{R}^c$, we obtain proposition 2.2.3. In particular, the condition on the Jacobian being non-zero is the same as requiring that $(f_*)_p$ to be surjective.

Proposition 2.4.4 Let $f : M \to N$ be a smooth map of manifolds such that $V = f^{-1}(c)$ is a regular level set. Then

$$T_p(V) = \ker((f_*)_p)$$

for any $p \in V$.

Proof Let $X_{\gamma,p} \in T_p(V)$. Then we have that

$$\begin{aligned}
(f_*)_p(X_{\gamma,p})(g) &= X_{\gamma,p}(g \circ f) \\
&= \frac{d(g \circ f \circ \gamma)}{dt}\Big|_{t=0} \\
&= \frac{dg(q)}{dt}\Big|_{t=0} \\
&= 0
\end{aligned}$$

for any choice of $g \in C^\infty_{N,f(p)}$ so that $T_p(V) \subseteq \ker((f_*)_p)$. Since c is a regular value, $(f_*)_p$ is surjective and so the rank of the linear map is $n = \dim(N)$. By rank nullity theorem, the kernel $\ker((f_*)_p)$ has dimension $m - n$. The preimage theorem also tells us that $T_p(V)$ has dimension $m - n$ and so we are done. ■

Theorem 2.4.5 (Sard's Theorem) Let M and N be smooth manifolds. Let $f : M \to N$ be a smooth map. Then the set of critical values of f has measure 0 in N.

2.5 Rank of the Differntial

The rank of the differential gives crucial information to how smooth maps of manifolds give rise to submanifolds. Consider a smooth map $f : M \to N$ between smooth manifolds. Let (U, ϕ) and $(V, \psi = (y^1, \dots, y^n))$ be charts on p and $f(p)$ respectively. We have seen that the map $(f_*)_p$ is given by the Jacobian $\left(\frac{\partial(y^i \circ \circ f \circ \phi^{-1})}{\partial x^j}\Big|_p\right)_{n\times m}$. It is easy to see that

$$(f_*)_p \text{ is injective} \quad\iff\quad m \leq n \text{ and } \operatorname{rank}\left(\left(\frac{\partial(y^i \circ \circ f \circ \phi^{-1})}{\partial x^j}\Big|_{\phi(p)}\right)_{n\times m}\right) = m$$

$$(f_*)_p \text{ is surjective} \quad\iff\quad m \geq n \text{ and } \operatorname{rank}\left(\left(\frac{\partial(y^i \circ \circ f \circ \phi^{-1})}{\partial x^j}\Big|_{\phi(p)}\right)_{n\times m}\right) = n$$

Theorem 2.5.1 (Immersion Theorem) Let $f : M \to N$ be a smooth map of smooth manifolds where $\dim(M) = m$ and $\dim(N) = n$. If $(f_*)_p$ is injective, then there are charts (U, ϕ) of M and (V, ψ) of N such that

$$(\psi \circ f \circ \phi^{-1})(x_1, \dots, x_m) = (x_1, \dots, x_m, 0, \dots, 0)$$

In other words, locally at each point, f is an inclusion $\mathbb{R}^m \to \mathbb{R}^n$.

Proof Suppose that $(f_*)_p$ is injective. Let (U, ϕ) be a chart on p and $(V, \psi = y_1, \dots, y_n)$ a chart on $f(p)$. Using proposition 3.2.2 gives us the Jacobian of $\psi \circ f \circ \phi^{-1}$. Since $(f_*)_p$ is injective, we have that the matrix has rank m. This means that we can choose indices $i_1, \dots, i_m$ from $\{1, \dots, m\}$ so that the submatrix with rows $i_1, \dots, i_m$, calling it A, is invertible. By the inverse function theorem, A gives a local diffeomorphism. Notice that functions $(y_{i_1} \circ f, \dots, y_{i_m} \circ f)$ is a function from U to $\mathbb{R}^m$. Since it is also diffeomorphic, it is actually a chart on M. This completes the proof. ■

Theorem 2.5.2 (Submersion Theorem) Let $f : M \to N$ be a smooth map of smooth manifolds where $\dim(M) = m$ and $\dim(N) = n$. If $(f_*)_p$ is surjective, then there are charts (U, ϕ) of M and (V, ψ) of N such that

$$(\psi \circ f \circ \phi^{-1})(x_1, \dots, x_n, x_{n+1}, \dots, x_m) = (x_1, \dots, x_n)$$

In other words, locally at each point, f is a projection $\mathbb{R}^m \to \mathbb{R}^n$.

Proof Suppose that $(f_*)_p$ is surjective. Let (U, ϕ) and (V, ψ) be charts around $p \in M$ and $f(p) \in N$ respectively. Then we know that f_* is given by the Jacobian matrix of $\psi \circ f \circ \phi^{-1}$. Since $(f_*)_p$ is surjective the $n \times m$ Jacobian matrix has rank n. This means that we can choose n columns from the $n \times m$ matrix for which the submatrix A of these n columns are invertible. By the inverse function theorem, A gives a local diffeomorphism. ■

Corollary 2.5.3 Let $f : M \to N$ be a smooth map of smooth manifolds where $\dim(M) = m$ and $\dim(N) = n$. If $(f_*)_p$ is bijective, then there are charts (U, ϕ) and (V, ψ) of p and $f(p)$ respectively such that $\psi \circ f \circ \phi^{-1} : \phi(U) \subseteq \mathbb{R}^m \to \psi(V) \subseteq \mathbb{R}^n$ is a homeomorphism. In particular, this means that $m = n$.

Proof Combine the immersion theorem and submersion theorem for the result. ■

We give names to the case when $(f_*)_p$ is injective or surjective for all points p on the manifold. In particular, this means that the immersion / submersion theorem applies to all points on the manifold.

Definition 2.5.4 (Immersions and Submersions) Let $f : M \to N$ be a smooth map of manifolds.

- If $(f_*)_p$ is injective for all $p \in M$, then f is called an immersion. We then say that M is immersed in N by f.
- If $(f_*)_p$ is surjective for all $p \in M$, then f is called an submersion.

Note that the definition of submersions and regular values are slightly different. Namely regular values are defined for points in the codomain while the condition on submersion checks every point on the domain of the smooth map.

Definition 2.5.5 (Embeddings) Let $f : M \to N$ be a smooth map of manifolds. We say that f is an embedding of M into N if f is injective, is an immersion and is homeomorphic to its image $f(M)$.

Proposition 2.5.6 Let M be a smooth manifold. Let N be a subset of M. Then N is an embedded submanifold of M if and only if the inclusion map $\iota : N \to M$ is an embedding.

Note that the condition on homeomorphism is necessary for $f(M)$ to be an embedded submanifold of N in the definition of embeddings only when M is not compact. This can be seen with the example of embedding a line to a torus. Consider

$$\varphi : \mathbb{R} \to L \subseteq \mathbb{R}^2 \to \frac{\mathbb{R}^2}{\mathbb{Z}^2} \cong T^2$$

where L is a line passing throught the origin of $\mathbb{R}^2$ with irrational slope. Then $\varphi(\mathbb{R})$ is dense in the torus T^2.

3 Some Examples of Smooth Manifolds

3.1 Manifolds in the Set of Real Matrices

Proposition 3.1.1 Let $n \in \mathbb{N} \setminus \{0\}$. Then the following are smooth submanifolds of $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$.

- The general linear group $GL(n, \mathbb{R})$
- The orthogonal group $O(n, \mathbb{R})$
- The special linear group $SL(n, \mathbb{R})$
- The special orthogonal group $SO(n, \mathbb{R})$.

Proof

- Notice that the determinant map $M_n(\mathbb{R}) \to \mathbb{R}$ defined by $A \mapsto \det(A)$ is smooth. Indeed the determinant is a degree n polynomial in each entry of an $n \times n$ matrix and polynomials $\mathbb{R}^2 \to \mathbb{R}$ are smooth. Then $GL(n, \mathbb{R})$ is precisely the set $\det^{-1}(\mathbb{R} \setminus \{0\})$ so that $GL(n, \mathbb{R})$ is an open subset of $M_n(\mathbb{R})$. By prp3.1.3, $GL(n, \mathbb{R})$ is a smooth manifold of dimension n^2.

■

Proposition 3.1.2 Let $n \in \mathbb{N} \setminus \{0\}$. Then the following are embedded submanifolds of $GL(n, \mathbb{R})$.

- The orthogonal group $O(n, \mathbb{R})$, with dimension $n^2 - (n^2 + n)/2 = (n^2 - n)/2$.
- The special linear group $SL(n, \mathbb{R})$
- The special orthogonal group $SO(n, \mathbb{R})$

Proof We first consider $O(n, \mathbb{R})$. Let S_n denote the set of symmetric matrices in $M_n(\mathbb{R})$. Since S is a finite dimensional vector space over $\mathbb{R}$, it is also a smooth manifold of the same dimension. Let $F : M_n(\mathbb{R}) \to S_n$ be the map $M \mapsto M^T M$. Notice that $O(n, \mathbb{R}) = F^{-1}(I_n)$. Let $A \in F^{-1}(I_n) = O(n)$. We can calculate the differential $(dF_*)_A : \mathbb{R}^{n^2} \cong M_n(\mathbb{R}) \to S_n \cong \mathbb{R}^{(n^2+n)/2}$ (vector space isomorphisms) by

$$(dF_*)_A(X) = \frac{d}{dt} F(A + tX)\bigg|_{t=0} = \frac{d}{dt}(A + tX)^T (A + tX)\bigg|_{t=0} = A^T X + X^T A$$

for any $X \in M_n(\mathbb{R})$. I claim that this linear map is surjective. Let $B \in S_n$ be a symmetric matrix. Using $X = \frac{1}{2} AB$, we have that

$$(dF_*)_A(AB) = A^T (\frac{1}{2} AB) + (\frac{1}{2} AB)^T A = \frac{1}{2}(B + B^T) = B$$

because A is orthogonal and B is symmetric. The preimage theorem is satisfied, hence $O(n, \mathbb{R})$ is an embedded submanifold of $GL(n, \mathbb{R})$ of dimension $\dim(O(n, \mathbb{R})) = n^2 - (n^2 + n)/2 = (n^2 - n)/2$.

■

4 Vector Bundles on Smooth Manifolds

While the previous section introduced general vector bundles over an arbitrary topological space, the concept of smooth bundles are exclusive solely to smooth manifolds. Indeed the notion of smoothness exists only in smooth manifolds when we say that the transition maps of the charts are smooth functions from Euclidean spaces.

4.1 Smooth Bundles and Smooth Sections

Definition 4.1.1 (Smooth Vector Bundles)
Let M be a smooth manifold. Let $p : E \to M$ be a vector bundle. We say that p is smooth if the following are true.

- E is a smooth manifold.
- p is a smooth map.
- The local trivialization charts are diffeomorphisms.

Proposition 4.1.2
Let M be a smooth manifold. Let $p_1 : E_1 \to M$ and $p_2 : E_2 \to M$ be smooth vector bundles. Then $E_1 \otimes E_2$ is a smooth bundle.

Proposition 4.1.3
Let M be a smooth manifold. Let $p : E \to M$ be a smooth vector bundle. Then E^* is a smooth bundle.

Definition 4.1.4 (Smooth Sections)
Let M be a smooth manifold. Let $p : E \to M$ be a smooth vector bundle. Define the smooth sections of p to be the subgroup

$$\Gamma^\infty(E) = \{s : M \to E \mid s \text{ is smooth and } p \circ s = \text{id}_M\} \leq \Gamma(E)$$

together with the operation $\mathcal{C}^\infty(M) \times \Gamma^\infty(E) \to \Gamma^\infty(E)$ given by $(f, s) \mapsto fs$.

Proposition 4.1.5
Let M be a smooth manifold. Let $p : E \to M$ be a smooth vector bundle. Then $\Gamma^\infty(E)$ is a $\mathcal{C}^\infty(M)$-module.

Proof
It is clear that $_1 + s_2$ and fs is a section. It remains to check that they are smooth sections. Let $p \in M$. Let (U, ϕ) be a smooth local trivialization of $p : E \to M$ containing p. Suppose that $\phi \circ s : M \to U \times \mathbb{R}^k$ is given by

$$(\phi \circ s_1)(q) = (q, a_1(q), \dots, a_k(q))$$

and similarly $(\phi \circ s_2)(q) = (q, b_1(q), \dots, b_k(q))$ for $q \in U$. Since s_1, s_2 are smooth maps, a_i and b_j are smooth functions on U. Since ϕ is linear on each fiber,

$$(\phi \circ (s_1 + s_2))(q) = (q, (a_1 + b_1)(q), \dots, (a_k + b_k)(q))$$

so that $s_1 + s_2$ is a smooth map on U and hence at p. Since p is an arbitrary point we conclude.

The proof for the module structure is similar to the above. ■

Definition 4.1.6 (Smooth Frames)
Let M be a smooth manifold. Let $p : E \to M$ be a smooth vector bundle. Let $(U, s_1, \dots, s_r)$ be a local frame of E. We say that the frame is smooth if $s_1, \dots, s_r \in \Gamma^\infty(E)$.

We write $\overline{e}_1, \dots, \overline{e}_r : M \to M \times \mathbb{R}^r$ for the trivial bundle $\pi : M \times \mathbb{R}^r \to M$, where each $\overline{e}_i : M \to M \times \mathbb{R}^r$ are defined by

$$\overline{e}_i(p) = (p, e_i)$$

for e_i the standard ith basis element of R^r. In particular, for a local trivialization $\phi : \pi^{-1}(U) \to U \times \mathbb{R}^r$ of a general vector bundle, we can define a smooth frame over the local trivialization in terms of frame of the trivial bundle. This is the frame $t_1, \dots, t_r : U \to E$ defined by $t_i(p) = \phi^{-1}(\overline{e_i}(p)) = \phi^{-1}(p, e_i)$ for each i.

4.2 Tensor Fields

Definition 4.2.1 (Tensor Fields)
Let M be a smooth manifold. Let $p : E \to M$ be a smooth vector bundle. A (p, q)-tensor field is an element of

$$\mathcal{T}_q^p(E) = \Gamma^\infty(E^{\otimes p} \otimes (E^*)^{\otimes q})$$

Lemma 4.2.2 (The Tensor Characterization Lemma)
Let M be a smooth manifold. Let $p : E \to M$ and $p_i : E_i \to M$ be smooth vector bundles for $1 \leq i \leq n$. Then there is an isomorphism of $\Gamma^\infty(M)$-modules

$$\text{Hom}_{\mathcal{C}^\infty(M)}\left(\prod_{i=1}^n \Gamma^\infty(E_i), \Gamma^\infty(E)\right) \cong \Gamma^\infty\left(\bigotimes_{i=1}^n E_i^* \otimes E\right)$$

For example, a $\mathcal{C}^\infty(M)$-linear map of sections $\Gamma^\infty(E_1) \to \Gamma^\infty(E_2)$ is the same data as as an element of $\Gamma^\infty(E_1^* \otimes E_2)$.

Definition 4.2.3 (Components of a Tensor Field)
Let M be a smooth manifold. Let $p : E \to M$ be a smooth vector bundle. Let F be a (p, q)-tensor field. Let $(U, s_1, \dots, s_k)$ be a smooth frame of E. Suppose that F is given locally at the frame by

$$F = \sum_{1 \leq i_1, \dots, i_p, j_1, \dots, j_q \leq k} F_{j_1 \cdots j_q}^{i_1 \cdots i_p} s_{i_1} \otimes \cdots \otimes s_{i_p} \otimes s_{j_1}^* \otimes \cdots \otimes s_{j_q}^*$$

We say that the components of F are the functions $F_{j_1 \cdots j_q}^{i_1 \cdots i_p} \in \mathcal{C}^\infty(U)$.

For a (p, q)-tensor field, there are n^{p+q} components in total.

Definition 4.2.4 (Contraction)
Let M be a smooth manifold. Let $p : E \to M$ be a smooth vector bundle of rank r. Let F be a tensor field of type (p, q). Define the contraction of F to be the $(p-1, q-1)$ tensor field $c_k^l(F)$ whose components are given by

$$c_k^l(F)_{j_1, \dots, j_{q-1}}^{i_1, \dots, i_{p-1}} = \sum_{t=1}^r F_{j_1, \dots, j_{l-1}, t, j_l, \dots, j_{q-1}}^{i_1, \dots, i_{k-1}, t, i_k, \dots, i_{p-1}}$$

In vector space notation, c_k^l has the effect of pairing the kth copy of V together with the lth copy of V^* with the canonical pairing $V \times V^* \to \mathbb{R}$ which is evaluation. Since we assume V^* comes with the dual basis of V, the canonical pairing always return 1.

4.3 The Tangent Bundle

Definition 4.3.1 (The Tangent Bundle)
Let M be a smooth manifold. Define the tangent bundle of M to be the disjoint union

$$TM = \coprod_{p \in M} T_p M$$

together with the projection map $\pi : TM \to M$ given by $\pi(v) = p$ if $v \in T_pM$.

Definition 4.3.2 (Local Trivialization Maps of the Tangent Bundle)
Let M be a smooth manifold. Let $(U, x^1, \dots, x^n)$ be a local chart of M. Define the induced local trivialization chart of TM to be the pair $(U, \widetilde{\phi})$ where the map $\widetilde{\phi} : TU = \pi^{-1}(U) \to U \times \mathbb{R}^n$ is given by

$$\widetilde{\phi}(v) = (\pi(v), c_1(p), \dots, c_n(p))$$

where $v \in T_pM$ is given locally at the chart by $v = \sum_{k=1}^n c_k(p) \frac{\partial}{\partial x^k}\Big|_p$.

Theorem 4.3.3 (Topology on the Tangent Bundle)
Let M be a smooth manifold of dimension n. Then TM is a topological space with basis

$$\mathcal{B} = \bigcup_\alpha \left\{ A \subseteq TM \;\middle|\; A \text{ is open in } T(U_\alpha) \text{ and } U_\alpha \text{ is a coordinate open set in } M \right\}$$

where we say that A is open in $T(U_\alpha)$ if and only if $\tilde{\phi}(A)$ is open in $\phi(U) \times \mathbb{R}^n \subseteq \mathbb{R}^{2n}$.

Proof
Firstly, notice that if $V \subseteq U$ is open, then $\phi(V) \times \mathbb{R}^n$ is an open subset of $\phi(U) \times \mathbb{R}^n$ so that the relative topology on $T(V)$ as a subset of TU is the same as the topology induced from the bijection $\tilde{\phi}|_{T(V)} : T(V) \to \phi(V) \times \mathbb{R}^n$.

Let $\{(U_\alpha, \phi_\alpha\}$ be the maximal atlas for M. Then

$$TM = \bigcup_\alpha T(U_\alpha) \subseteq \bigcup_{A \in \mathcal{B}} A \subseteq TM$$

so that equality holds everywhere.

Let A be open in $T(U)$ and B open in $T(V)$. The goal is to show that $A \cap B$ is open in $T(U \cap V)$. Since $T(U \cap V)$ is a subspace of $T(U)$, by definition of relative topology, $A \cap T(U \cap V)$ is open in $T(U \cap V)$. Similarly, $B \cap T(U \cap V)$ is open in $T(U \cap V)$ so that

$$A \cap B \subseteq T(U) \cap T(V) = T(U \cap V)$$

is open in $T(U \cap V)$. ∎

Proposition 4.3.4
Let M be a smooth manifold of dimension n. Then the tangent bundle TM is a vector bundle of dimension n.

Proof

∎

Proposition 4.3.5
Let M be a manifold. Then the tangent space TM is a manifold of dimension $2n$.

Proof Let $\{(U_\alpha, \phi_\alpha\}$ be the maximal atlas for M and $\mathcal{B}$ a countable basis for M. For each coordinate open set U_α and $p \in U_\alpha$, choose a basic open set $B_{p,\alpha} \in \mathcal{B}$ such that $p \in B_{p,\alpha} \subset U_\alpha$. The collection $\{B_{p,\alpha}\}$ without duplicate elements is a subcollection of $\mathcal{B}$ and is countable. For any open set U in M and $p \in U$, there is a coordinate open set U_α such that $p \in U_\alpha \subset U$. Hence $p \in B_{p,\alpha} \subset U$ so that $\{B_{p,\alpha}\}$ is a countable basis for M. Let $\{U_i \mid i \in \mathbb{N}\}$ be a countable basis of M consisting of coordinate open sets. Let ϕ_i be the coordinate map on U_i. Since $T(U_i)$ is homeomorphic to the open subset $\phi_i(U_i) \times \mathbb{R}^n \subseteq \mathbb{R}^{2n}$ and any subset of a Euclidean space is second countable, $T(U_i)$ is second countable. For each i, choose a countable basis $\{B_{i,j} \mid j \in \mathbb{N}\}$ for $T(U_i)$. Then $\{B_{i,j} \mid i,j \in \mathbb{N}\}$ is a countable basis for $T(M)$ so that (TM) is second countable.

For Hausdorffness,

Finally, if $\{(U_\alpha, \phi_\alpha\}$ is a smooth atlas on M, then $\{(T(U_\alpha), \widetilde{\phi_\alpha})\}$ is a smooth atlas for $T(M)$. It is clear that $T(M) = \bigcup_\alpha T(U_\alpha)$. It remains to check that they are smoothly compatible. Recall that by proposition 3.1.7, if $(U_\alpha, \phi_\alpha = (x^1, \dots, x^n))$ and $(U_\beta, \phi_\beta = (y^1, \dots, y^n))$ are charts on M and $p \in U \cap V$ such that

$$X_p = \sum_{k=1}^n a_k(p) \frac{\partial}{\partial x^k}\bigg|_p = \sum_{k=1}^n b_k(p) \frac{\partial}{\partial y^k}\bigg|_p$$

are the two representations of the same tangent vector in the two charts, then we have

$$a_i(p) = \sum_{j=1}^n b_j(p) \frac{\partial x^i}{\partial y^j}\bigg|_p \quad \text{and} \quad b_i(p) = \sum_{k=1}^n a_k(p) \frac{\partial y^i}{\partial x^k}\bigg|_p$$

Now we have

$$\widetilde{\phi_\beta} \circ \widetilde{\phi_\alpha}^{-1} : \phi_\alpha(U_\alpha \cap U_\beta) \times \mathbb{R}^n \to \phi_\beta(U_\alpha \cap U_\beta) \times \mathbb{R}^n$$

is given by

$$(\phi_\alpha(p), a_1(p), \dots, a_n(p)) \mapsto \left(p, \sum_{k=1}^n a_k(p) \frac{\partial}{\partial x^k}\bigg|_p\right) \mapsto (\phi_\beta(p), b_1(p), \dots, b_n(p))$$

where each $b_i(p)$ is given by the formula

$$b_i(p) = \sum_{k=1}^n a_k(p) \frac{\partial y^i}{\partial x^k}\bigg|_p = \sum_{k=1}^n a_k(p) \frac{\partial(\phi_\beta \circ \phi_\alpha^{-1})^i}{\partial x_j}\bigg|_{\phi_\alpha(p)}$$

where $(\phi_\beta \circ \phi_\alpha^{-1})^i$ means the ith component of the map $\phi_\beta \circ \phi_\alpha^{-1}$. Therefore $\widetilde{\phi_\beta} \circ \widetilde{\phi_\alpha}^{-1}$ is smooth. Thus we conclude. ■

Proposition 4.3.6
Let M be a smooth manifold. Let F be a tensor field of type (p,q) over TM. Let $(U, x^1, \dots, x^n)$ be a local chart of M. Then the components of F is given by

$$F^{i_1 \cdots i_p}_{j_1 \cdots j_q} = F\left(dx^{i_1}, \dots, dx^{i_p}, \frac{\partial}{\partial x^{j_1}}, \dots, \frac{\partial}{\partial x^{j_q}}\right)$$

4.4 Vector Fields

Definition 4.4.1 (Smooth Vector Fields)
Let M be a smooth manifold. Define the smooth vector fields of M to be the $\mathcal{C}^\infty(M)$-module

$$\mathfrak{X}(M) = \Gamma^\infty(TM)$$

Definition 4.4.2 (Parallelizable Manifolds)
Let M be a smooth manifold. We say that M is parallelizable if TM is isomorphic to the trivial bundle.

From topology, we know that M is parallelizable if and only if M admits $\dim(M)$ amount vector fields that form a basis when restricted to each fiber.

Example 4.4.3
S^1 is parallelizable.

Proof

■

4.5 Integral Curves of Vector Fields

Definition 4.5.1 (Integral Curves)
Let M be a smooth manifold. Let $X \in \mathfrak{X}(M)$ be a smooth vector field. Let $\gamma : I \subseteq \mathbb{R} \to M$ be a smooth curve. We say that γ is an integral curve of X if

$$d\gamma\left(\frac{d}{dt}\bigg|_{t=t_0}\right) = X_{\gamma(t_0)}$$

for all $t_0 \in I$.

Example 4.5.2
Let M be a smooth manifold. Let $X \in \mathfrak{X}(M)$ be a smooth vector field. Let $\gamma : I \subseteq \mathbb{R} \to M$ be a smooth curve. Suppose that $(U, x^1, \dots, x^n)$ is a chart at M such that $\text{im}(\gamma) \subseteq U$. Then γ is an integral curve of X if and only if

$$\frac{d(x^i \circ \gamma)}{dt} = a_i \circ \gamma$$

This is just a classical system of first order ODEs.

Proposition 4.5.3
Let M be a smooth manifold. Let $X \in \mathfrak{X}(M)$ be a smooth vector field. Let $p \in M$. Then there exists an open interval $I_p \subseteq \mathbb{R}$ containing 0 and a unique smooth curve $\gamma_p : I_p \subseteq \mathbb{R} \to M$ such that the following are true.

- $\gamma_p(0) = p$.
- γ_p is an integral curve of X.

Moreover, the map $(t, p) \mapsto \gamma_p(t)$ is smooth.

Definition 4.5.4 (Maximal Integral Curves)
Let M be a smooth manifold. Let $X \in \mathfrak{X}(M)$ be a smooth vector field. Let $p \in M$. Let $\gamma : I \subseteq \mathbb{R} \to M$ be an integral curve of X. We say that γ is maximal through p if the following are true.

- $\gamma(0) = p$.
- For any other integral curve $\phi : J \subseteq \mathbb{R} \to M$, we have $J \subseteq I$.

We denote the domain of γ by $I = D(p)$.

Proposition 4.5.5
Let M be a smooth manifold. Let $X \in \mathfrak{X}(M)$ be a smooth vector field. Then integral flows of X exists and is unique.

Definition 4.5.6 (The Flow of a Vector Field)
Let M be a smooth manifold. Let $X \in \mathfrak{X}(M)$ be a smooth vector field.

- Define the flow domain of X to be

$$D^X = \{(t, p) \in \mathbb{R} \times M \mid t \in D(p)\}$$

- Define the flow of X to be the map $\theta : D^X \to M$ given by

$$\theta^X(t, p) = \gamma_p(t)$$

where γ_p is the maximal integral curve through p.

Proposition 4.5.7
Let M be a smooth manifold. Let $X \in \mathfrak{X}(M)$ be a smooth vector field. Then the following are true.
- For all $p \in M$, we have $\theta^X(0, p) = p$.
- For all $(s, p) \in D^X$ and $(t, \theta^X(s, p)) \in D^X$, we have
$$\theta^X(t, \theta^X(s, p)) = \theta^X(s + t, p)$$
- The map $\theta^X : D^X \to M$ is smooth.
- For each t, the map $\theta^X(t, -)$ is a diffeomorphism with inverse $\theta^X(-t, -)$.

Definition 4.5.8 (Complete Vector Fields)
Let M be a smooth manifold. Let $X \in \mathfrak{X}(M)$ be a smooth vector field. We say that X is complete if $D^X = \mathbb{R} \times M$.

Proposition 4.5.9
Let M be a smooth manifold. Let $X \in \mathfrak{X}(M)$ be a smooth vector field. If the set
$$\overline{\{p \in M \mid X_p \neq 0\}}$$
is compact, then X is complete.

Proof
For every point p, let $U(p)$ be a neighbourhood of p and $\varepsilon(p) > 0$ such that there exists a local 1-parameter group of local diffeomorphisms Φ defined on $I_{\varepsilon(p)} \times U(p)$ inducing X. Since M is compact, finitely many of the $U(p)$, say $U(p_1), \dots, U(p_k)$ cover M. Set $\varepsilon = \min\{\varepsilon(p_1), \dots, \varepsilon(p_k)\}$. We see that $\Phi_t(p)$ can be defined on $I_\varepsilon \times M$, hence on all of $\mathbb{R} \times M$. ■

Corollary 4.5.10
If M is a smooth compact manifold, then every smooth vector field on M is complete.

4.6 One Parameter Family of Diffeomorphisms

Definition 4.6.1 (One Parameter Family of Diffeomorphisms)
Let M be a smooth manifold. Let $\varphi : \mathbb{R} \times M \to M$ be a smooth map. We say that φ is a one parameter family of diffeomorphisms if the following are true.
- For each $t \in \mathbb{R}$, we have $\varphi_t : M \to M$ is a diffeomorphism.
- $\varphi(0, -) = \mathrm{id}_M$.
- $\varphi(s + t, -) = \varphi(t, -) \circ \varphi(s, -)$ for all $s, t \in \mathbb{R}$.

Example 4.6.2
Let M be a smooth manifold. Let $X \in \mathfrak{X}(M)$ be a complete vector field. Then θ^X is a one parameter family of diffeomorphisms.

Definition 4.6.3 (Infinitesimal Generator)
Let M be a smooth manifold. Let φ be a one parameter family of diffeomorphisms. Define the infinitesimal generator of φ to be the vector field $X_\varphi \in \mathfrak{X}(M)$ given by
$$X_\varphi = \frac{d}{dt}\bigg|_{t=0} \varphi(t, -)$$

Proposition 4.6.4
Let M be a smooth manifold. Then there is a one to one bijection

$$\{X\in\mathfrak{X}(M)\mid X\text{ is complete }\}\quad\overset{1:1}{\longleftrightarrow}\quad\{\varphi:\mathbb{R}\times M\to M\mid\varphi\text{ is a one parameter family of diffeomorphisms }\}$$

5 Differential Forms and its Applications

In considering a smooth section on the tangent bundle we obtain the notion of smooth vector fields. The smooth section on the cotangent bundle will lead to smooth 1-forms. However, the theory of k-forms also provide a notion of integration on manifolds. In particular, Stoke's theorem on n-forms generalizes all notions of divergence and curl, including Green's theorem and the divergence theorem. By considering all differential k-forms as a whole, they also contain topological information including the number of holes of space. This odd similarity with that of cohomology in Algebraic Topology leads to de Rham's theorem, which states that there is an isomorphism between singular cohomology of a smooth manifold and de Rham cohomology.

5.1 Differential 1-Forms

Definition 5.1.1 (Differential 1-Forms)
Let M be a smooth manifold. Define the differential forms of M to be the $\mathcal{C}^\infty(M)$-module

$$\Omega^1(M) = \Gamma^\infty(T^*M)$$

Definition 5.1.2 (Differential of a Function) Let M be a smooth manifold. For each $f \in C^\infty(M)$, define $df : M \to T^*(M)$ to be the map given by

$$(df)_p(X_p) = X_p(f)$$

for each $p \in M$.

Note that the above definition makes sense. Firstly, df is a map from M so to each point p on M, we assign a covector $(df)_p$. But a covector is none other than an $\mathbb{R}$-linear map from $T_p(M)$ to $\mathbb{R}$ so we need to define what $(df)_p$ does to each $X_p \in T_p(M)$.

Recall from linear algebra that V and its dual space V^* has the same dimension if V is finite dimensional. This applies also to the case $T_p^*(M)$. Moreover, we can compute a basis for $T_p^*(M)$ in terms of differential of some particular functions in $C^\infty(M)$, at least locally.

Proposition 5.1.3 Let M be a smooth manifold. Let $(U, \phi = (x^1, \dots, x^n))$ be a chart at $p \in M$. Write dx^k the differential of the smooth function $x^k : U \to \mathbb{R}$ (so that it is a 1-form). Then

$$\{(dx^1)_p, \dots, (dx^n)_p\}$$

is the dual basis in $T_p^*(M)$ of $\left\{ \frac{\partial}{\partial x^1}\bigg|_p, \dots, \frac{\partial}{\partial x^n}\bigg|_p \right\}$.

Proof By definition, we have that

$$(dx^j)_p \left(\frac{\partial}{\partial x^i}\bigg|_p \right) = \frac{\partial x^j}{\partial x^i}\bigg|_p = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

and so we conclude. ■

Proposition 5.1.4
Let M be a smooth manifold. Let $f \in \mathcal{C}^\infty(M)$ be a smooth function. Let $(U, x^1, \dots, x^n)$ be a chart on M. Then locally on the chart,

$$(df)_p = \sum_{k=1}^n \frac{\partial f}{\partial x^k}\bigg|_p (dx^k)_p$$

for $p \in U$.

Proof
Since $dx^1, \dots, dx^n$ is a smooth local frame for T^*M, there exists smooth functions $a_1, \dots, a_n : U \to \mathbb{R}$ such that

$$df = \sum_{k=1}^n a_k dx^k$$

Evaluating $\frac{\partial}{\partial x^i}$ on both sides give

$$df\left(\frac{\partial}{\partial x^i}\right) = \sum_{k=1}^n a_k dx^k \left(\frac{\partial}{\partial x^i}\right)$$
$$\frac{\partial f}{\partial x^i} = a_i$$

so we are done. ■

Lemma 5.1.5
Let M be a smooth manifold. Then the map $d : C^\infty(M) \to \Omega^1(M)$ is an $\mathbb{R}$-derivation.

Proof
Follows from the fact that $\frac{\partial}{\partial x^i}$ is an $\mathbb{R}$-derivation. ■

5.2 Differential k-Forms

Recall that $\Lambda(T^*(M))$ is a graded algebra that enjoys addition and wedging as multiplication.

Definition 5.2.1 (Differential k-Forms)
Let M be a smooth manifold. Define the differential k-forms of M to be the $C^\infty(M)$-module

$$\Omega^k(M) = \Gamma^\infty(\Lambda^k(T^*M))$$

Locally on a chart $(U, x^1, \dots, x^n)$ of M, a differential k-form ω is given by

$$\omega = \sum_{1 \leq i_1 < \cdots < i_k \leq n} a_{i_1, \dots, i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k}$$

Proposition 5.2.2
Let M be a smooth manifold. Let $\omega \in \Omega^k(M)$ and $\tau \in \Omega^l(M)$. Then $\omega \wedge \tau \in \Omega^{k+l}(M)$.

Proof We know that $\omega \wedge \tau \in \Lambda^{k+l}(T(M))$ by construction of the exterior product. Explicitly: if $\omega = \sum f_I dx^I$ and $\tau = \sum g_J dx^J$, then

$$\omega \wedge \tau = \sum f_I g_J dx^I \wedge dx^J$$

Clearly if f_I and g_J are smooth, then their product is also smooth. ■

Definition 5.2.3 (The Graded Ring of Differential Forms)
Let M be a smooth manifold. Define the differential forms of M to be the graded $C^\infty(M)$-module

$$\Omega^\bullet(M) = \bigoplus_{i=0}^{\dim(M)} \Omega^i(M)$$

together with multiplication given by $\wedge : \Omega^k(M) \times \Omega^l(M) \to \Omega^{k+l}(M)$.

5.3 The Induced Pullback Map

Now let $f : M \to N$ be a smooth map of manifolds. Recall that for each $p \in M$, there is a pushforward map of tangent spaces: $(f_*)_p : T_pM \to T_{f(p)}N$. Taking dual of the tangent spaces, we obtain the dual map $(f_*)_p^* : T_{f(p)}^*N \to T_p^*M$. We can then take the exterior power of the map to obtain

$$\Lambda^k((f_*)_p^*) : \Lambda^k(T_{f(p)}^*N) \to \Lambda^k(T_p^*M)$$

which is the main object of the next definition. Denote this map by f^*.

Definition 5.3.1 (The Pullback Map)
Let M, N be smooth manifolds. Let $f : M \to N$ be a smooth map. Define the pullback map of f on differential forms as follows. For $\omega \in \Omega^k(N)$, the map $f^* : \Omega^\bullet(N) \to \Omega^\bullet(M)$ is defined pointwise as

$$f^*(\omega_{f(p)}) = \Lambda^k((f_*)_p^*)(\omega_p)$$

Proposition 5.3.2
Let M, N be smooth manifolds. Let $f : M \to N$ be a smooth map. Let $\omega \in \Omega^k(N)$ be a smooth k-form on N. Let $(V, y^1, \dots, y^n)$ be a chart on N. Suppose that the components of ω at the chart is given by $a_{i_1,\dots,i_k} : V \to \mathbb{R}$. Then locally at a chart containing $f^{-1}(V)$, we have

$$f^*(\omega) = \sum_{1 \le i_1 < \dots < i_k \le n} (a_{i_1,\dots,i_k} \circ f) f^*(dy^{i_1}) \wedge \dots \wedge f^*(dy^{i_k})$$

Proposition 5.3.3
Let M, N be smooth manifolds. Let $f : M \to N$ be a smooth map. Let $(U, \phi = (x^1, \dots, x^m))$ and $(V, \psi = (y^1, \dots, y^n))$ be local charts on M and N respectively such that $f(U) \subseteq V$. If f is a diffeomorphism, then $n = m$ and we have

$$f^*(dy^1 \wedge \dots \wedge dy^n) = \det(f_*)\, dx^1 \wedge \dots \wedge dx^n$$

where f_* is the Jacobian.

Proposition 5.3.4
Let M, N be smooth manifolds. Let $f : M \to N$ be a smooth map. Then $f^* : \Omega^\bullet(N) \to \Omega^\bullet(M)$ is an $\mathbb{R}$-algebra homomorphism.

Proposition 5.3.5
The following are true regarding the pullback map.

- Let M, N, K be smooth manifolds. Let $f : M \to N$ and $g : N \to K$ be smooth maps. Then we have $(g \circ f)^* = f^* \circ g^*$.
- Let M be a smooth manifold. Then we have $(\mathrm{id}_M)^* = \mathrm{id}_{\Omega^\bullet(M)}$.

5.4 Exterior Derivatives

Definition 5.4.1 (Exterior Derivative)
Let M be a smooth manifold. An exterior derivative on M is a map $d : \Omega^\bullet(M) \to \Omega^\bullet(M)$ such that the following are true.

- $d(\Omega^k(M)) \subseteq \Omega^{k+1}(M)$ is an $\mathbb{R}$-linear map
- $d \circ d = 0$
- If $\omega \in \Omega^r(M)$ and $\tau \in \Omega^s(M)$ then $d(\omega \wedge \tau) = d\omega \wedge \tau + (-1)^r \omega \wedge d\tau$
- For any $f \in C^\infty(M)$, df is the differential of f as defined above.

Proposition 5.4.2
Let M be a smooth manifold. Then the exterior derivative exists and is unique. Moreover, if (U, ϕ)

is a local chart on M and $\omega = \sum_I a_I dx^I \in \Omega^k(M)$ is a differential k-form, then locally on the chart

$$d\omega = \sum_I (da_I) \wedge dx^I = \sum_I \sum_{k=1}^n \frac{\partial a_I}{\partial x^k} dx^k \wedge dx^I$$

Proof Locally on each chart define $d : \Omega^\bullet(M) \to \Omega^\bullet(M)$ by the formula given. Then using a partition of unity subordinate to the open cover of M via the charts, patch it up globally. Now we check the above criteria for the exterior derivative.

- Differentiation in $\mathbb{R}^n$ is $\mathbb{R}$-linear. The exterior derivative d is defined as differentiation on local charts of M so that d is also $\mathbb{R}$-linear. By definition of the total differential, $d(\omega) \in \Omega^{k+1}(M)$ for $\omega \in \Omega^k(M)$.
- For $\omega \in \Omega^k(M)$, we have that

$$d^2\omega = \sum_I \sum_{j=1}^n \sum_{i=1}^n \frac{\partial^2 a_I}{\partial x^j \partial x^i} \, dx^i \wedge dx^j \wedge dx^I$$

  Notice that for $i < j$, there are two copies of $\frac{\partial^2 a_I}{\partial x^j \partial x^i}$ in the triple sum since partial differentiation commutes while $dx^i \wedge dx^j = -dx^j \wedge dx^i$ by property of the exterior algebra. Thus the terms cancel out and we obtain $d^2\omega = 0$.
- It suffices to prove it for expressions of the form $\omega = f dx^I$ and $\tau = g dx^J$. By $\mathbb{R}$-linearity this extends to the rest of $\Omega^\bullet(M)$. We have that

$$\begin{aligned} d(\omega \wedge \tau) &= d(fg \; dx^I \wedge dx^J) \\ &= (fdg + gdf) \wedge dx^I \wedge dx^J \\ &= (-1)^k f \; dx^I \wedge g \wedge dx^J + df \; \wedge dx^I \wedge g dx^J \\ &= d\omega \wedge \tau + (-1)^k \omega \wedge d\tau \end{aligned}$$

  proving the claim.
- It is clear that f is a 0-form then df is the differential of f by the given equation.

It remains to show that the definition of d is invariant under the choice of chart so that the partition of unity can patch it up. Suppose that $(U, \phi = (x^1, \dots, x^n))$ and $(V, \psi = (y^1, \dots, y^n))$ are two charts on M such that

$$\omega = \sum_I f_I \; dx^I \quad \text{and} \quad \omega = \sum_I g_I \; dy^I$$

on (U, ϕ) and (V, ψ) respectively. Suppose that on (U, ϕ) we have the exterior derivative d and d' on (V, ψ) respectively. Then on $U \cap V$, we have

$$d\omega = d\left(\sum_I g_I \; dy^I\right) = \sum_I d(g_I) \; dy^I + \sum_I g_I \; d(dy^I)$$

But we have that

$$\begin{aligned} d(dy^I) &= d^2 y^I - dy^{i_1} \wedge d(dy^{i_2} \wedge \dots \wedge dy^{i_k}) \\ &= -dy^{i_1} \wedge d(dy^{i_2} \wedge \dots \wedge dy^{i_k}) \\ &\vdots \\ &= 0 \end{aligned}$$

so that $d\omega = \sum_I d(g_I) \; dy^I = d'\omega$ and we conclude. ■

Proposition 5.4.3 Let $f : M \to N$ be a smooth map of smooth manifolds. Let $\omega \in \Omega^k(N)$. Then

$$d(f^*\omega) = f^*(d\omega)$$

Proof It suffices to prove this on a local chart. Let $(V, \psi = (y^1, \dots, y^n))$ be a chart of N. We first prove the case for a 0-form. Let h be a 0-form. It suffices to show this for every p in the chart. Let $X_p \in T_p(N)$. Then on one hand, we have that

$$d(f^*h)_p(X_p) = X_p(f^*(h)) = X_p(h \circ f)$$

On the other hand, we have that

$$\begin{aligned} (f^*((dh)_p))(X_p) &= (dh)_{f(p)}(f^*(X_p)) && \text{(Pull back of a one form)} \\ &= (f^*(X_p))(h) && \text{(Definition of the differntial)} \\ &= X_p(h \circ f) && \text{(Definition of } f^*\text{)} \end{aligned}$$

and so we are done.

Let ω be a smooth k-form. ω is given locally by $\omega = \sum_I a_I dy^I$ for some $a_I : V \to \mathbb{R}$ that is smooth. Then we have that $f^*(\omega) = \sum_I (a_I \circ f)\, df_{i_1} \wedge df i_k$ so that

$$d(f^*(\omega)) = \sum_I d(a_I \circ f) \wedge df_{i_1} \wedge df i_k$$

On the other hand, we have that

$$\begin{aligned} f^*(d\omega) &= f^*\left(\sum_I d(a_I) \wedge dy^I\right) \\ &= \sum_I d(a_I \circ f) \wedge df_{i_1} \wedge df i_k \end{aligned}$$

and so we are done. ■

5.5 Orientation of Smooth Manifold

Definition 5.5.1 (Orientation)
Let M be a smooth manifold of dimension n. An orientation of M is an equivalence class $[\omega] \in \frac{\Omega^n(M)}{\sim}$ of smooth n-forms where we say that $\omega_1 \sim \omega_2$ if there exists an everywhere positive smooth function $f \in \mathcal{C}^\infty(M)$ such that $\omega_1 = f\omega_2$.

Definition 5.5.2 (Orientable Manifolds)
Let M be a smooth manifold. We say that M is orientable if M has an orientation.

Definition 5.5.3 (Oriented Atlas)
Let M be a smooth orientable manifold. Let $\{(U_i, \phi_i) \mid i \in I\}$ be an atlas on M. We say that the atlas is oriented if for any two charts $(U, \phi = (x^1, \dots, x^n))$ and $(V, y^1, \dots, y^n)$ in the atlas we have that

$$\det\left(\left(\frac{\partial y^j \circ \phi^{-1}}{\partial x_i}\right)_{1 \le i,j \le n}\right) > 0$$

Proposition 5.5.4
Let M be a smooth manifold. Then M is orientable if and only if M admits an oriented atlas.

Proof Suppose that M is orientable. This means that we have a nowhere vanishing top form

$$\omega = f\; dx^1 \wedge \cdots \wedge dx^n$$

in a coordinate chart $(U, \phi = (x^1, \dots, x^n))$ on M. By replacing $x^1 : U \to \mathbb{R}$ with $-x^1 : U \to \mathbb{R}$, we can assume that M is covered by such coordinate charts where the function corresponding to f above is positive in each chart. Suppose that $(V, \psi = (y^1, \dots, y^n))$ is another chart on M with non-empty intersection with U. Locally on V, suppose that ω takes the form of

$$\omega = g\; dy^1 \wedge \cdots \wedge dy^n$$

Then on $U \cap V$, we have that

$$\begin{aligned} g\; dy^1 \wedge \cdots \wedge dy^n &= (g \circ \phi^{-1})\; \det \left(\frac{\partial (y^i \circ \phi^{-1})}{\partial x^j} \right)\; dx^1 \wedge \cdots \wedge dx^n \\ &= f\; dx^1 \wedge \cdots \wedge dx^n \end{aligned}$$

So if $f > 0$ and $g > 0$, we must have that the determinant greater than 0 and so we are done.

Conversely, suppose that the determinant is non zero on all intersections of a covering $\{U_\alpha \mid \alpha \in i\}$ of M. Choose a partition of unity $\{\theta_\alpha \mid \alpha \in I\}$ subordinate to the covering. Define a top form on M by

$$\omega = \sum_{\alpha \in I} \theta_\alpha\; dy_\alpha^1 \wedge \cdots \wedge dy_\alpha^n$$

which is smooth since θ_α is smooth for each $\alpha \in I$. Let $(U_\beta, \psi = (y^1, \dots, y^n))$ be a a coordinate chart in the covering. Then locally we have that

$$\omega|_{U_\beta} = \sum_{i \in I} \theta_\alpha \det \left(\frac{\partial y_\alpha^i}{\partial x_\beta^j} \right)\; dy_\beta^1 \wedge \cdots \wedge dy_\beta^n$$

The form ω is non-negative since $\theta_\alpha > 0$ and the determinant is positive by assumption. It is non-zero since at each point, at least one of the functions θ_α is non-zero, as they all sum to 1. Thus we are done. ■

It is easy to see that a choice of orientation is a choice of coordinate charts such that either the nowhere vanishing form $\omega = f\; dx^1 \wedge \cdots \wedge dx^n$ has its coefficient f positive or negative throughout all coordinate charts. Because this way, the determinant of the transition functions will always have positive determinant.

Proposition 5.5.5 A connected manifold M has exactly two orientations.

Proposition 5.5.6 If a manifold with boundary M is orientable, then ∂M is also orientable.

Proof If $\dim(M) = 1$, then $\dim(\partial M) = 0$ and consists of points so that the result is trivial. Let $n = \dim(M) \geq 2$. Choose a local chart $(U, \phi = (x^1, \dots, x^n))$ such that ∂M is defined by $x_n = 0$ locally and on intersections with other charts $\det \left(\left(\frac{\partial y^i}{\partial x^j} \right)_{1 \leq i,j \leq n} \right) > 0$ Where two local charts with coordinates $(U, \phi = (x^1, \dots, x^n))$ and $(V, \psi = (y^1, \dots, y^n))$ intersect. Since $x^n = 0$ and

$y^n = 0$, the Jacobian matrix for a point $p \in \partial M$ is given by

$$\begin{pmatrix} \frac{\partial y^1}{\partial x^1}\Big|_p & \cdots & \frac{\partial y^1}{\partial x^{n-1}}\Big|_p & \frac{\partial y^1}{\partial x^n}\Big|_p \\ \vdots & \ddots & \vdots & \vdots \\ \frac{\partial y^{n-1}}{\partial x^1}\Big|_p & \cdots & \frac{\partial y^{n-1}}{\partial x^{n-1}}\Big|_p & \vdots \\ 0 & \cdots & 0 & \frac{\partial y^n}{\partial x^n}\Big|_p \end{pmatrix}$$

Now $\psi \circ \phi^{-1}$ maps $x_n > 0$ to $y_n > 0$. Thus if $x_n = 0$, then $y_n = 0$ and if $x_n > 0$, then $y_n > 0$. Thus

$$\frac{\partial y^n}{\partial x^n}\Big|_{x_n = 0} > 0$$

and since the determinant of the above matrix is positive, the determinant of the upper left $(n-1) \times (n-1)$ matrix also has positive determinant. Thus $(U_\alpha|_{\partial M}, \phi|_\alpha|_{\partial M})$ give an atlas for ∂M whose transition functions have everywhere positive Jacobian. Thus ∂M is orientable. ∎

In order to complete Stoke's theorem, we need a canonical orientation on the boundary on the manifold that depends on the choice of orientation on the manifold itself.

Definition 5.5.7 (Induced Orientation) Let M be a smooth orientable manifold with boundary of dimension n. Let ω be a nowhere vanishing 1-form representing an orientation on M. Suppose that in local coordinates, ω on the boundary of M is given by

$$\omega = \epsilon \, dx^1 \wedge \cdots \wedge dx^n$$

where $\epsilon = \pm 1$ and $x^n > 0$. Then define the induced orientation ω' on ∂M locally by

$$\omega' = (-1)^n \epsilon \, dx^1 \wedge \cdots \wedge dx^{n-1}$$

5.6 Integration of Top Forms on Orientable Manifolds

There are a number of conditions for integration on manifolds. Namely,

1. The manifold must be oriented
2. Only top forms can be integrated
3. The top forms must have compact support.

Once the above conditions are satisfied, we can use the partition of unity to define integration.

Definition 5.6.1 (Compactly Supported Differential Forms)
Let M be a smooth manifold. Let $\omega \in \Omega^k(M)$ be a differential form of M. We say that ω has compact support if

$$\overline{\{p \in M \mid \omega_p \neq 0\}} \subseteq M$$

is compact.

Definition 5.6.2 (Integration of Smooth Top Forms)
Let M be an oriented smooth manifold of dimension n. Let $\omega \in \Omega^n(M)$ be a compactly supported smooth top form. Define the integral of ω over M as follows. Choose an oriented atlas $\{(U_i, \phi_i) \mid i \in I\}$ of M. Choose a partition of unity $\{\theta_i \mid i \in I\}$ subordinate to the open cover of M such that $\omega = \sum_{i \in I} \theta_i \omega$. Suppose that locally on each chart, we have $\theta_i \omega|_{U_i} = g_i dx_i^1 \wedge \cdots \wedge dx_i^n$

where each $g_i \in \mathcal{C}^\infty(M)$ has compact support. Then define the integral of ω over M by

$$\int_M \omega = \sum_{i\in I}\int_{\phi_i(U_i)} g_i \circ \phi_i^{-1}$$

where the integral on the right is the Riemann integral in $\mathbb{R}^n$.

One property of partition of unity is that only finitely many θ_α are not identically 0 on ω, so the above sum is finite. Moreover, since ω has compact support, each $g_\alpha \circ \phi_\alpha^{-1}$ is also of compact support. It remains to show that the definition of integral is independent of the choice of the partition of unity and the choice of the open cover.

Proposition 5.6.3 Let M be an orientable manifold of dimension n. Let $\omega \in \Omega^n(M)$ be a top form of compact support. Then $\int_M \omega$ is independent of the choice of the open cover and partition of unity.

Proposition 5.6.4 Let M be an oriented manifold of dimension n without boundary. Let $\omega \in \Omega^n(M)$ be a top form of M with compact support. If $-M$ denotes the opposite choice of orientation of M, then

$$\int_{-M} \omega = -\int_M \omega$$

Proposition 5.6.5
Let M be an n-dimensional oriented manifold. Let ω be an $n-1$ form with compact support. Then

$$\int_M d\omega = 0$$

Proof
Choose a covering of M and a partition of unity $\{\theta_i \mid i \in I\}$ subordinate to the cover such that we can write

$$\omega = \sum_{i\in I}\theta_i\omega$$

Then in a chart, we can write

$$\theta_i\omega = f_1\, dx^2\wedge\cdots\wedge dx^n - f_2\, dx^1\wedge dx^3\wedge\cdots\wedge dx^n + \cdots + (-1)^{n+1} f_n\, dx^1\wedge\cdots\wedge dx^{n-1}$$

so that

$$d(\theta_i\omega) = \left(\frac{\partial f_1}{\partial x_1} + \cdots + \frac{\partial f_n}{\partial x_n}\right) dx^1\wedge\cdots\wedge dx^n$$

Now notice that

$$\int_{\mathbb{R}^n} \frac{\partial f_1}{\partial x_1}\, dx^1\wedge\cdots\wedge dx^n = \int_{\mathbb{R}}\int_{\mathbb{R}}\cdots\left(\int_{\mathbb{R}}\frac{\partial f_1}{\partial x_1}\, dx^1\right) dx^2\cdots dx^n$$

using Fubini's theorem. Since f_1 has compact support, it vanishes for $|x_1| > N$ for some $N \in \mathbb{R}$. So

$$\int_{\mathbb{R}} \frac{\partial f_1}{\partial x_1}\, dx^1 = f_1(N) - f_1(-N) = 0$$

as N tends to infinity. Continuing in the same way, the entire integral evaluates to 0 and so we conclude. ■

This theorem in context says that integration on the manifold that does not touch the boundary contributes to the overall integration a sum of 0. Indeed we are assuming M to be an ordinary manifold without boundary in the above theorem.

Theorem 5.6.6 (Stokes' Theorem)
Let M be an n-dimensional oriented manifold with boundary ∂M carrying the induced orientation. Let ω be an $n-1$ form with compact support. Then

$$\int_M d\omega = \int_{\partial M} \omega|_{\partial M}$$

Proof
Notice that when $n=1$, it is entirely the same as the fundamental theorem of calculus so that we only treat the case for when $n \geq 2$. Write $\omega = \sum_{i \in I} \theta_i \omega$ where $\{\theta_i \mid i \in I\}$ is a partition of unity subordinate to a choice of open cover on M. Now we have that

$$\int_M d\omega = \sum_{i \in I} \int_M d(\theta_i \omega)$$

and we can write

$$\theta_i \omega = f_1 \, dx^2 \wedge \cdots \wedge dx^n - f_2 \, dx^1 \wedge dx^3 \wedge \cdots \wedge dx^n + \cdots + (-1)^{n+1} f_n \, dx^1 \wedge \cdots \wedge dx^{n-1}$$

similar to the above proof. There are two types of chart in the choice of open cover. If U_α does not intersect with the boundary, the above theorem implies that the contribution to the integral is 0. For those that do intersect, we have that

$$\begin{aligned} \int_M d(\theta_i \omega) &= \int_{x_n \geq 0} \left(\frac{\partial f_1}{\partial x_1} + \cdots + \frac{\partial f_n}{\partial x_n} \right) \, dx_1 \cdots \, dx_{n-1} \\ &= \int_{\mathbb{R}^{n-1}} [f_n]_0^\infty \, dx_1 \cdots dx_{n-1} \\ &= -\int_{\mathbb{R}^{n-1}} f_n(x_1, \dots, x_{n-1}, 0) \, dx_1 \cdots dx_{n-1} \\ &= \int_{\partial M} \theta_i \omega|_{\partial M} \end{aligned}$$

where the last equality follows because

$$\theta_i \omega|_{\partial M} = (-1)^{n-1} f_n \, dx^1 \wedge \cdots dx^{n-1}$$

and we used the induced orientation $(-1)^n \epsilon dx^1 \wedge \cdots \wedge dx^{n-1}$. ■

6 De Rham Cohomology

6.1 The Cohomology Groups

By proposition 6.2.4, the operator $d : \Omega^k(M) \to \Omega^{k+1}(M)$ for a smooth manifold M is in fact a boundary operator. Thus we can define a cochain complex using differential forms.

Definition 6.1.1 (De Rham Complex) Let M be a smooth manifold of dimension n. Define the de Rham complex of M to be the cochain complex

$$0 \longrightarrow \Omega^0(M) = C^\infty(M) \xrightarrow{d} \Omega^1(M) \longrightarrow \cdots \longrightarrow \Omega^{n-1}(M) \xrightarrow{d} \Omega^n(M) \longrightarrow 0$$

The complex is denoted as $(\Omega^\bullet(M), d)$, coinciding with the algebra of smooth differential forms.

Definition 6.1.2 (Closed and Exact Forms) Let α be a k-form of $\mathbb{R}^n$. Then α is closed if $d\alpha = 0$, and α is exact if there exists a $k-1$-form such that $d\beta = \alpha$.

The reason for these names is because in algebraic topology, we can define the cohomology group of this chain of algebras (called a complex). In particular, it is defined by taking the closed k-forms and quotienting the exact k-forms.

Definition 6.1.3 (de Rham Cohomology Groups) Let M be a smooth manifold of dimension n. Define the de Rham cohomology groups of M to be the cohomology of the chain of differential forms:

$$H^k_{\text{dR}}(M) = H^k(\Omega^\bullet(M); \mathbb{R})$$

for $0 \leq k \leq n$.

Proposition 6.1.4 Let M be a smooth manifold of dimension n. Then the following are true for the de Rham cohomology of M.

- $H^k_{\text{dR}}(M)$ is a vector space over $\mathbb{R}$ for all $k \in \mathbb{N}$.
- For $r > n$ we have $H^r_{\text{dR}}(M) = 0$
- If M has k connected components then $H^0_{\text{dR}}(M) \cong \mathbb{R}^k$

Definition 6.1.5 (de Rham Cohomology Ring)
Let M be a smooth manifold. Define the de Rham cohomology ring to be the direct sum

$$H^*_{\text{dR}}(M) = \bigoplus_{i=0}^{\dim(M)} H^i_{\text{dR}}(M)$$

together with multiplication given by exterior products.

Check: H^*_{dR} is a graded $\mathbb{R}$-algebra.

Proposition 6.1.6
Let M, N be smooth manifolds. Let $f : M \to N$ be a smooth map. Then the induced map

$$f^* : H^*_{\text{dR}}(N) \to H^*_{\text{dR}}(M)$$

given by $[\omega] \mapsto [f^*\omega]$ is a well defined $\mathbb{R}$-algebra homomorphism.

6.2 Relation to Orientability

Proposition 6.2.1
Let M be a smooth manifold. Then the following are true.

- M is orientable if and only if $H^{\dim(M)}_{\text{dR}}(M) \cong \mathbb{R}$.
- M is not orientable if and onlyif $H^{\dim(M)}_{\text{dR}}(M) = 0$.

Proposition 6.2.2
Let M be a compact orientable connected smooth manifold. Then the map

$$\int_M : H_{\text{dR}}^{\dim(M)}(M) \to \mathbb{R}$$

is a natural isomorphism.

7 Integration of Densities

7.1 The Density Bundle

Definition 7.1.1 (Densities of a Vector Space)
Let V be a finite dimensional vector space over $\mathbb{R}$. Let $\mu : V^{\dim(V)} \to \mathbb{R}$ be a function. We say that μ is a density of V if for any linear map $T : V \to V$, we have

$$\mu(T(v_1), \dots, T(v_n)) = |\det(T)|\mu(v_1, \dots, v_n)$$

Definition 7.1.2 (The Space of Densities)
Let V be a finite dimensional vector space over $\mathbb{R}$. Define the space of densities of V to be tje set

$$\text{Den}(V) = \{\mu : V^{\dim(V)} \to \mathbb{R} \mid \mu \text{ is a density } \}$$

together with addition and scalar multiplication.

Lemma 7.1.3
Let V be a finite dimensional vector space over $\mathbb{R}$. Then $\text{Den}(V)$ is a vector space over $\mathbb{R}$ of dimension 1.

Definition 7.1.4 (The Density Bundle)
Let M be a smooth manifold. Define the density bundle of M by the set

$$\text{Den}(M) = \coprod_{p \in M} \text{Den}(T_pM)$$

together with charts given as follows. For $(U, x^1, \dots, x^n)$ a local chart of M, the function $\left|dx^1 \wedge \cdots \wedge dx^n\right|$ is a chart of $\text{Den}(M)$.

Definition 7.1.5 (Densities)
Let M be a smooth manifold. A density of M is an element

$$\mu \in \mathcal{C}^\infty(M, \text{Den}(M))$$

Definition 7.1.6 (The Absolute Value)
Let M be a smooth manifold of dimension n. Define the map $|\cdot| : \Lambda^n T^*M \to \text{Den}(M)$ as follows. Let $\omega \in \Lambda^n T^*M$. Let $(U, x^1, \dots, x^n)$ be a local chart of M. Suppose that ω is given locally by $\omega = f dx^1 \wedge \cdots \wedge dx^n$. Define $|\omega|$ to be the density whose locally at the same chart is given by $|f|\left|dx^1 \wedge \cdots \wedge dx^n\right|$.

Proposition 7.1.7
Let M be a smooth manifold. Then the following are true.
- $|\cdot| : \Lambda^n T^*M \to \text{Den}(M)$ is a bundle homomorphism.
- If M is orientable, then $|\cdot|$ is an isomorphism.

Definition 7.1.8 (Pullback of Densities)
Let M, N be smooth manifolds of dimension n. Let $F : M \to N$ be a smooth map. Let $\mu : N \to \text{Den}(N)$ be a density of N. Define the pullback of μ by F to be the density $F^*\mu \in \mathcal{C}^\infty(M, \text{Den}(M))$ given by

$$F^*\mu_p(v_1, \dots, v_n) = \mu_{F(p)}(dF_p(v_1), \dots, dF_p(v_n))$$

Proposition 7.1.9
Let M, N be smooth manifolds of dimension n. Let $F : M \to N$ be a smooth map. Let μ :

$N \to \text{Den}(N)$ be a density of N. Let $(V, y^1, \dots, y^n)$ be a local chart of N. Let $(U, x^1, \dots, x^n)$ be a local chart of M such that $F^{-1}(V) \subseteq U$. Suppose that μ is given locally at the chart by $\mu = u\big|dy^1 \wedge \cdots \wedge dy^n\big|$. Then locally at the charts, we have

$$F^*(u\big|dy^1 \wedge \cdots \wedge dy^n\big|) = (u \circ F)|\det(DF)|\big|dx^1 \wedge \cdots \wedge dx^n\big|$$

7.2 Integration of Densities

Definition 7.2.1 (Compactly Supported Densities)
Let M be a smooth manifold. Let $\mu \in \mathcal{C}^\infty(M, \text{Den}(M))$ be a density of M. We say that μ has compact support if

$$\overline{\{p \in M \mid \mu_p \neq 0\}} \subseteq M$$

is compact.

Definition 7.2.2 (Integration of Densities)
Let M be a smooth manifold of dimension n. Let $\mu \in \mathcal{C}^\infty(M, \text{Den}(M))$ be a compactly supported density of M. Define the integral of ω over M as follows. Choose an atlas $\{(U_i, \phi_i) \mid i \in I\}$ of M. Choose a partition of unity $\{\theta_i \mid i \in I\}$ subordinate to the open cover of M such that $\mu = \sum_{i\in I} \theta_i \mu$. Suppose that locally on each chart, we have $\theta_i \mu|_{U_i} = g_i dx_i^1 \wedge \cdots \wedge dx_i^n$ where each $g_i \in \mathcal{C}^\infty(M)$ has compact support. Then define the integral of μ over M by

$$\int_M \mu = \sum_{i\in I} \int_{\phi_i(U_i)} g_i \circ \phi_i^{-1}$$

where the integral on the right is the Riemann integral in $\mathbb{R}^n$.